

Total No. of Questions : 10]

SEAT No. :

P3243

[5461]-281

[Total No. of Pages : 2

B. E. (I.T.)

INFORMATION AND CYBER SECURITY
(2012 Course) (Semester - I) (End Sem.) (414453)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) What is the purpose of S-boxes in DES? Explain the steps in detail. [6]

b) Calculate using Euclidean algorithm [4]

GCD (55,22)

GCD (1970, 1066)

OR

Q2) a) Describe the steps to generate a Digital certificate. [6]

b) Describe the different types of cryptanalytic attacks based on what is known to the attacker. [4]

Q3) a) Describe any two modes of operation (ECB, CBC, CFB, OFB) with the help of diagram. [6]

b) Distinguish between symmetric and asymmetric cipher. [4]

OR

Q4) a) Calculate the shared symmetric key using Diffie Hellman algorithm. [6]

$a = 7, q = 11, X_a = 3, X_b = 6$

b) What are the steps to generate and verify a Digital signature. [4]

Q5) a) List and state the purpose of different protocols of SSL. What are the steps in SSL Record protocol? [8]

b) Explain IDS and its types? [8]

OR

P.T.O.

Q6) a) What services are provided by AH and ESP. List and describe the different fields of AH and ESP header. [8]

b) Discuss : [8]

i) IKE protocol

ii) Transport vs Tunnel mode

Q7) a) What is cyberstalking? Explain its working. [8]

b) Discuss Indian legal perspective on cybercrimes. [8]

OR

Q8) a) Describe any two cybercrimes and their respective penalties. [8]

b) How social engineering plays wide role in cybercrime? Discuss with the help of example. [8]

Q9) Write a short note on following [18]

a) DOS and DDOS

b) Key-loggers and Spywares

c) Phishing

OR

Q10) Write a short note on following [18]

a) Proxy servers and Anonymizers

b) Viruses and Worms

c) SQL injection

